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Stacked Polyphase Bridge Converters for High-Voltage Traction Drives: A Review and System-Level Design Perspectives

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ABSTRACT Stacked polyphase bridge (SPB) converters partition a high dc-link voltage among series-connected three-phase bridges, each supplying an electrically separated machine winding group. This architecture enables lower-voltage semiconductor devices and modular converter–machine integration, but requires coordinated modulation, voltage balancing, and fault management. This paper reviews SPB topology evolution, operating principles, machine integration, semiconductor scaling, and control strategies for high-voltage traction drives. It synthesizes the effects of carrier interleaving on dc-link ripple, the conditions and limitations of common-mode cancellation, and the balancing behavior of different control structures. The review connects cell count and device voltage class with winding configuration, gate-drive complexity, cooling, and post-fault operation. These interdependencies inform an iterative system-design framework, illustrated by a high-voltage traction example. The synthesis shows that semiconductor-level benefits must be evaluated alongside the additional converter and machine requirements, and that the preferred cell count depends on the application and operating mission. Joint battery–converter–machine design are identified as priorities for establishing the practical competitiveness of SPB traction drives.

INDEX TERMS Electric vehicles, integrated motor drives, modular converters, multiphase machines, stacked polyphase bridge (SPB) converters.

I. Introduction

Electrification is pushing traction-drive propulsion and fast-charging power toward several hundred kilowatts; meanwhile, system-level performance requirements continue to rise [1]. At that point, increasing the dc-link voltage has emerged as a primary design lever for meeting both demands. On the propulsion side, a higher dc-link voltage can improve efficiency and power density [2]. On the battery and charging side, it reduces the current required for a given power level, lowering cable conduction loss and conductor cross-section while enabling higher charging power under a given current limit [3].

This trend, however, is not without consequence at the converter level. A higher dc-link voltage also brings more severe insulation-coordination and electromagnetic-compatibility (EMC) demands, together with added thermal and reliability stress on the power stage [2], [3]. Raising the dc-link voltage therefore requires coordinated converter, machine, and thermal design.

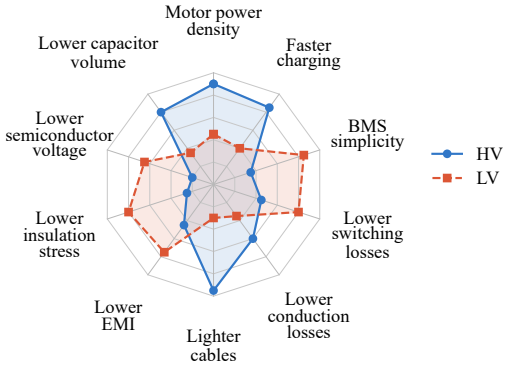


FIGURE 1. Conceptual illustration of qualitative system-level tradeoffs between low-voltage and high-voltage traction architectures.

Fig. 1 summarizes these qualitative tradeoffs between conventional low-voltage and higher-voltage traction architectures.

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TABLE 1. Representative voltage and power levels in published traction-inverter studies, industrial and reference platforms, and production battery-electric vehicles.

Published studies / demonstrators	Industrial / reference platforms	Production vehicles
Bertelshofer et al.: 400 V, 150 kW [6]	**Bosch Gen. 4**:400/800 V, 250/350 kW [7]	**BMW i4 M50**: 400 V, 400 kW [8]
Taha et al.: 800 V, 100 kW [9]	**Wolfspeed reference design**: 800 V, 300 kW [10]	**Hyundai IONIQ 5 N**: 800 V, 478 kW [11]
Al-Hmoud et al.: 1 kV, 200 kW [12]	**Infineon eCAV platform**: 800 V, 250 kW [13]	**Volvo EX90 Performance**: 800 V, 500 kW [14]
Wang et al.: 1.2 kV, 100 kW [15]	**AVL PI850e**: 800 V, 250 kW [16]	**Porsche Taycan Turbo S**: 800 V, 700 kW [17]
Ayaz et al.: 1.2 kV, 250 kW [18]	**Punch IV5**: 800 V, 600 kW [19]	**Lucid Air Dream Performance**: 900 V, 828 kW [20]

A. Literature and Industry Trends Toward Higher DC-Link Voltage

The trend motivated above is not merely conceptual: it is already established in academic prototypes, industrial inverter platforms, and production vehicles, as summarized in Table 1. Reported academic demonstrators range from 400-V to 800-V and 1.2-kV converters, with still higher levels investigated for high-power passenger and heavy-duty applications [3], while industrial platforms increasingly target 800-V operation as a baseline and production vehicles show a comparable shift from established 400-V architectures to a growing number of 800-V and higher-voltage platforms [4], [5]. This shift makes semiconductor blocking-voltage capability a central converter-design constraint.

B. Semiconductor Blocking-Voltage Tradeoff

In a conventional two-level voltage-source inverter, each switch must withstand the dc-link voltage. Increasing the blocking-voltage rating of a unipolar device generally increases its specific on-state resistance, linking system voltage to conduction loss [21], [22]. Switching performance also depends on output charge. Following [23], these effects are represented by

$$R_{\mathrm{on,sp}} = k_R U_B^{\alpha_R}, \qquad C_{\mathrm{oss,Q,sp}} = k_C U_B^{\alpha_C},$$

where U_B is the device blocking voltage and the empirical coefficients are taken from [23]. The resulting device figure-of-merit is

$$\mathrm{D{-}FOM} = \frac{1}{\sqrt{R_{\mathrm{on,sp}} C_{\mathrm{oss,Q,sp}}}} \propto U_B^{-(\alpha_R+\alpha_C)/2}.$$

Here, $C_{\mathrm{oss,Q}}(U) = Q_{\mathrm{oss}}(U)/U$ is the charge-equivalent capacitance at the evaluated switching voltage. The resistance–capacitance product is independent of die area; a higher D-FOM indicates lower minimum hard-switching semiconductor losses under the model assumptions.

Fig. 2 shows the fitted trends over the approximate voltage ranges of the source survey, with on-state resistance at 25 °C. Lower blocking voltage improves D-FOM within each fitted technology, but comparisons across technologies require consistent voltage and operating conditions. Wide-bandgap devices expand the available design choices [21], [24]–[26]; their system benefit remains dependent on device selection, switching conditions, and converter implementation. Voltage

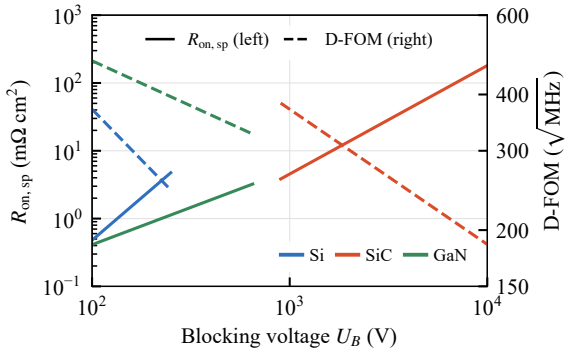


FIGURE 2. Specific on-state resistance and D-FOM versus blocking voltage, using empirical device fits from [23].

partitioning therefore offers a route to exploiting lower-voltage devices, subject to the additional hardware and control requirements of the topology.

C. Multilevel Converters and Their Limitations

Multilevel converters provide a way to partition voltage and utilize lower-voltage devices. NPC, T-type, flying-capacitor, and cascaded architectures all distribute the dc-link voltage among additional semiconductor positions or capacitor-voltage levels, reducing individual device voltage stress [27]–[30].

They require additional semiconductor positions, passive components, and gate drivers, which increases the number of potential failure points. Classical three-phase multilevel bridges still supply a common winding set, whereas cascaded and other modular architectures can provide module-level redundancy. In either case, fault tolerance depends on available isolation paths, spare voltage capability, and control; component count alone establishes neither fault tolerance nor system reliability.

D. Stacked Polyphase Bridge

The stacked polyphase bridge (SPB) distributes the high-voltage conversion function among N repeated converter units. Each unit is a conventional three-phase two-level bridge connected in series on the dc side, supplying its own electrically separated three-phase winding group of the same machine, as illustrated in Fig. 3.

Under balanced operation, each cell's local dc-link voltage $V_{\mathrm{dc},k} \approx V_{\mathrm{dc}}/N$, so the semiconductor voltage rating can be

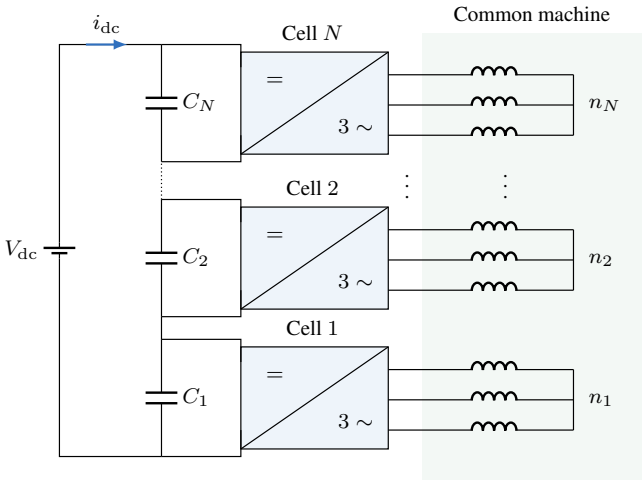


FIGURE 3. Conceptual architecture of an N-cell SPB traction drive: series-connected local dc-link capacitors, series-connected three-phase two-level bridge cells, and their electrically separated machine winding groups.

selected on the local cell voltage rather than the complete traction dc-link voltage – addressing the semiconductor bottleneck much as multilevel converters do. Each cell also drives a separated winding group, providing modularity at the converter–machine interface. A faulted cell can, in principle, be isolated and bypassed while the remaining cells continue to drive the machine at reduced power. It is this combination of semiconductor voltage reduction and converter–machine modularity that distinguishes SPB from conventional multilevel and cascaded architectures and motivates its consideration as a candidate topology for next-generation traction drives.

Realizing this potential, however, introduces coupled design challenges in component count, dc-link dynamics, machine design, common-mode voltage, and fault handling, addressed throughout this review. These challenges have grown more tractable as neighboring research areas have matured: multiphase machines provide additional post-fault degrees of freedom, integrated modular motor drives show how repeated power-electronic units can be physically associated with individual machine sections, wide-bandgap semiconductors amplify performance differences among voltage classes, and distributed control coordinates modular converter units [31]–[36].

Despite more than a decade of SPB research, beginning with Norrga *et al.* [37], the feasibility of electric-vehicle traction applications requires these developments to be assessed together. The available literature remains distributed across converter topology, modulation, dc-link stability, multiphase-machine design, integrated motor drives, semiconductor technology, thermal management, distributed control, and fault-tolerant operation. Although these topics are strongly coupled in an SPB traction drive, they are rarely evaluated within a common system boundary, leaving a gap between a well-motivated architectural concept and a system-level feasibility assessment. Accordingly, this paper provides a system-level review of SPB traction-drive technology aimed at closing that gap.

E. Review Scope and Contribution

This is a focused narrative review rather than an exhaustive systematic survey or a meta-analysis. Direct SPB evidence concerns series-connected dc ports supplying electrically separated winding groups; related multiphase and integrated-drive studies provide enabling methods but are not treated as demonstrations of SPB traction performance. The review distinguishes reported results, analytical illustrations, and application-level inferences. Its contribution is to connect device voltage partitioning to winding feasibility, modulation, balancing, common-mode excitation, and post-fault operation within one EV design boundary.

The remainder of the paper is organized as follows. First, SPB's historical development, operating principle, and modulation are consolidated. Second, the interaction between cell count and the electric machine is examined alongside multiphase-machine and integrated-modular-drive research. Third, common-mode voltage and dc-link dynamics and balancing are examined. Fourth, semiconductor voltage scaling and fault handling are treated as coupled architecture-level design problems. Finally, the remaining research gaps and automotive validation requirements are identified.

II. Evolution of the Stacked Polyphase Bridge Concept

SPB belongs to the broader development of modular converter–machine systems. Modular and integrated modular motor drives established repeated converter–winding units, machine segmentation, power sharing, and fault-tolerant operation [31], [38]. Series-connected implementations extended these concepts to dc-voltage sharing, enabling lower-voltage semiconductor devices [32], [39], [40]. Other modular drive topologies adopted the same principle [41], [42].

Norrga *et al.* introduced SPB in 2013 [37], followed by dedicated control and modulation methods [43]. Related series-connected drives demonstrated internal dc-link regulation using machine-side degrees of freedom [44]. Subsequent stability analyses established the balancing-controller requirements [45], [46], identifying dc-link balancing as a central control challenge alongside semiconductor voltage selection.

Research then expanded toward coordinated modulation, scalable control, and converter–machine design. Modulation studies linked switching loss to dc-link harmonics [47], while distributed control provided an alternative to centralized balancing [48]. Comparative assessment against parallel-connected polyphase bridges and other modular converters evaluated machine design, losses, capacitor energy, cost, and redundancy, positioning SPB among several candidate architectures [49]. Interleaving and phase-count studies further connected converter performance to machine architecture [34], [50]–[52].

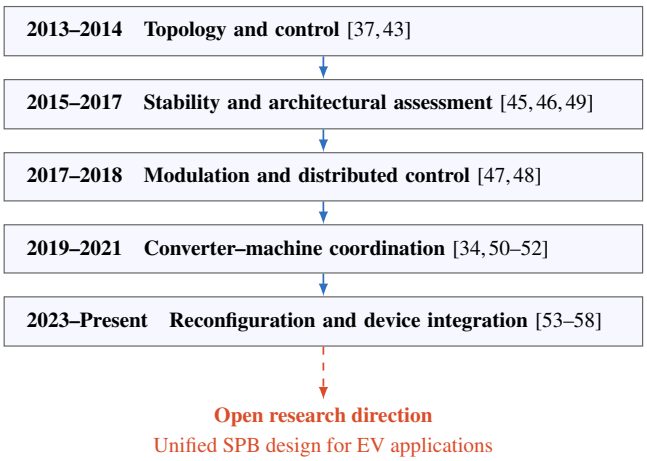


FIGURE 4. Representative developments in SPB research grouped by period and research focus.

Recent work has extended SPB toward decentralized and reconfigurable operation. Common-duty-ratio control, decentralized balancing, and multi-agent power distribution reduce reliance on centralized coordination [53]–[56]. Online module isolation and reintroduction enable dynamic dc-bus reconfiguration, including removal of a faulted cell without interrupting torque production [57]. In parallel, gallium nitride devices have renewed interest in the benefits of lower semiconductor voltage classes [58].

Fig. 4 groups these developments by research focus, tracing the progression from voltage partitioning to coordinated converter–machine design and reconfigurable operation. The remaining integration gap for electric vehicles is a unified, application-specific SPB design framework that couples battery and dc-link requirements, semiconductor selection, machine configuration, and control, evaluated against common energy, power-density, cost, and reliability targets.

III. Operating Principle and Topological Characteristics

A. Basic Structure and Voltage Partitioning

An N-cell SPB consists of N conventional three-phase two-level voltage-source inverter cells connected in series on the dc side, with each bridge supplying one electrically separated three-phase winding group of the same machine [37], [43], [47]. The machine therefore presents N ac ports, while the converter cells share a single series dc path.

The dc-link voltage satisfies

$$V_{\mathrm{dc}} = \sum_{k=1}^{N} V_{\mathrm{dc},k}, \qquad V_{\mathrm{dc},k} \approx \frac{V_{\mathrm{dc}}}{N} \tag{1}$$

under balanced operation. A basic implementation contains $6N$ active switches and N local dc-link capacitors. Increasing N reduces the nominal cell voltage, and hence the required semiconductor voltage class, but simultaneously increases the number of switches, gate drivers, sensors, capacitors, and machine-winding terminals that must be realized and managed. Fig. 5 shows this structure at switch level for two representative cells, making explicit the $6N$-switch bridge realization, the local capacitors C_k, and the notation used throughout the remainder of this paper.

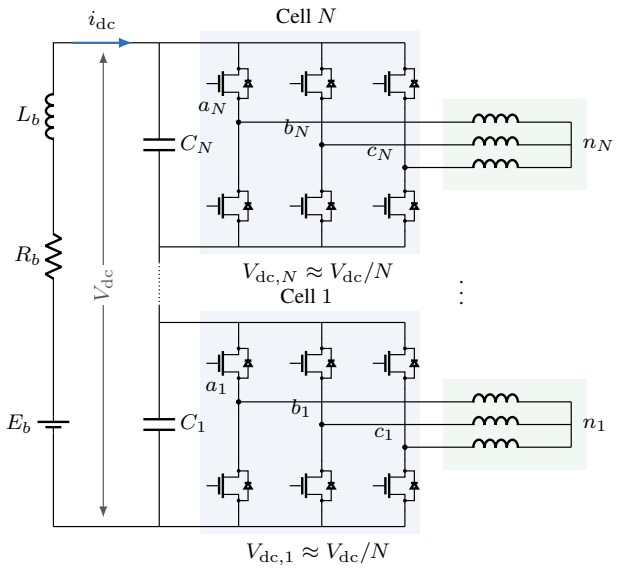


FIGURE 5. Switch-level realization of the SPB dc–ac stage for two representative cells (Cell 1 and Cell N; cells $2, \ldots, N-1$ elided).

B. Series DC Connection and Power Flow

Neglecting auxiliary and bypass paths, the cell dc currents are constrained to be equal,

$$i_{\mathrm{dc},1} = i_{\mathrm{dc},2} = \cdots = i_{\mathrm{dc},N} = i_{\mathrm{dc}}, \tag{2}$$

so that the dc-side power entering cell k is

$$p_{\mathrm{dc},k} = V_{\mathrm{dc},k} i_{\mathrm{dc}}. \tag{3}$$

The total machine power, by contrast, is the sum of the powers delivered to the individual winding groups,

$$P_{\mathrm{ac}} = \sum_{k=1}^{N} P_k. \tag{4}$$

Under symmetrical operation $P_k \approx P_{\mathrm{ac}}/N$, but equal power sharing is not a fundamental requirement of the topology. The multi-three-phase machine provides additional current and power degrees of freedom that allow power to be redistributed among winding groups [44], [55]. This redistribution capability is the physical basis for both dc-link voltage balancing and several of the post-fault operating strategies.

The common dc current and independently controlled ac powers make local capacitor voltages internal dynamic states. If one winding group processes more power than another, the corresponding capacitor energies can diverge even though all cells carry the same dc current [45], [46]. Managing this divergence is the dc-link voltage-balancing problem, SPB's first-order control challenge.

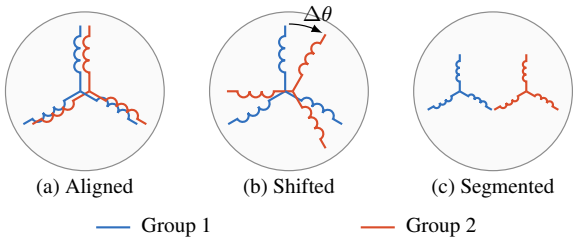


FIGURE 6. Schematic comparison of winding-group spatial arrangements: (a) aligned groups (parallel phase axes), (b) spatially shifted groups (offset by $\Delta\theta$), and (c) physically segmented stator modules.

C. Cell Count as a Converter–Machine Design Variable

Cell count is consequently a converter–machine design variable rather than a semiconductor variable alone. Increasing N also reduces the voltage available to each winding group and alters the number of winding sets, terminal count, electromagnetic coupling, current-control degrees of freedom, and available reconfiguration states.

Machine phase current, however, does not generally scale as $1/N$: it instead depends on the number of turns, flux linkage, winding connection, torque constant, and machine geometry. A many-cell architecture may therefore reduce the semiconductor voltage class while still requiring high-current devices or parallel chips at the device level.

IV. Electric-Machine Requirements and Integrated Implementation

A. Winding Architecture and Electromagnetic Coupling

Although the winding groups are electrically separated at their terminals, they share the same magnetic circuit, rotor, air gap, and shaft. Their electromagnetic coupling depends on winding distribution, slot/pole combination, spatial displacement, and machine geometry [34], [44], [55], [59]. Therefore, an SPB-compatible machine cannot generally be obtained simply by dividing a conventional three-phase winding into multiple groups without considering the resulting coupling and harmonic behavior.

The three-phase groups may be aligned, spatially shifted, or implemented as physically segmented stator modules. Spatial displacement affects mutual coupling, torque-ripple components, and the additional current subspaces available to the controller, among other effects [60]–[62]. These additional subspaces can introduce non-torque-producing currents, but they can also be exploited for harmonic suppression, differential power sharing, and fault-tolerant operation [55], [63]. Split winding architectures show a similar tradeoff: they can reduce ripple and provide additional current/voltage degrees of freedom, but they do so by increasing winding terminals and coupling complexity [64]. These three winding-group arrangements are compared schematically in Fig. 6.

For the kth PMSM winding group, a steady-state approximation neglecting mutual coupling between groups gives

$$\begin{aligned} v_{d,k} &= R_{s,k} i_{d,k} - \omega_{\mathrm{e}} L_{q,k} i_{q,k}, \\ v_{q,k} &= R_{s,k} i_{q,k} + \omega_{\mathrm{e}} \left(L_{d,k} i_{d,k} + \psi_{f,k} \right), \end{aligned} \tag{5}$$

For coupled or transient operation, the flux-linkage model must retain $\dot{\psi}_{d,k}$ and $\dot{\psi}_{q,k}$ and the dependence of flux on all winding-group currents. The required voltage must satisfy

$$\sqrt{v_{d,k}^2 + v_{q,k}^2} \le k_{\mathrm{mod}} V_{\mathrm{dc},k} \approx k_{\mathrm{mod}} \frac{V_{\mathrm{dc}}}{N}. \tag{6}$$

This falling per-cell voltage budget ties winding turns, inductance, and field-weakening capability directly to N – a coupling demonstrated in optimized multiphase SPB-PMSM systems [34]. Recent multiphase traction-machine studies similarly emphasize coordinated design of winding voltage, harmonic subspaces, and modulation [65], [66].

B. From Multiphase Machine to Integrated Modular Drive

Physical integration can convert the added electrical complexity of many winding groups into a packaging advantage. With a remote inverter, additional winding groups increase phase-cable and connector count; positioning each converter bridge close to its corresponding winding group instead reduces external cabling, shortens high-frequency interconnections, distributes heat sources, and aligns converter modules with stator segments.

This design philosophy predates SPB and has been extensively studied in integrated and modular motor drives [31], [32], [35], [39], [40], [67]–[79], summarized in Table 2.

C. Physical Integration Options

The electrically separated winding groups required by SPB do not impose a unique converter placement on a radial-flux machine. Chowdhury *et al.* [80], consistent with the broader integrated-motor-drive survey of Abebe *et al.* [81], classify motor/inverter physical integration into four categories: whether the converter mounts radially or axially, and whether it interfaces with a separate housing/end-cap structure or directly with the stator core. This taxonomy frames how SPB converter cells can be placed around a radial-flux machine, as illustrated in Fig. 7. The axial-flux stator-mounted and polygon-housing alternative is covered by the Mohamed *et al.* developments [75]–[77] in Table 2.

D. Integration Is Not Free of Complexity

Integration does not remove complexity; it relocates it. Converter modules mounted on or near the machine are exposed to elevated temperature, vibration, and sealing/coolant requirements, among other stresses, and to electromagnetic coupling between gate-drive, power, and machine conductors. Shared machine–converter cooling can also increase thermal interaction between the stator and power electronics [75].

SPB introduces an additional insulation challenge because its converter cells and associated winding groups occupy

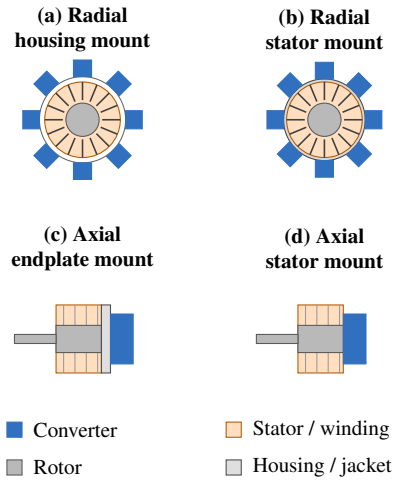


FIGURE 7. Physical integration options for a radial-flux machine, following [80].

TABLE 2. Representative Integrated Motor-Drive Developments

Reference	Integration theme
Shakweh *et al.* [67]	Integrated motor/inverter architecture
Wheeler/Pickering *et al.* [68], [69]	Converter and thermal design inside motor envelope
Brown *et al.* [70]	Converter mounted at modular stator pole
Wolmarans *et al.* [71]	Fault-tolerant converter, machine, and thermal design
Choi *et al.* [31]	Modular PM machine and high-temperature electronics
Wang *et al.* [39], [40]	Low-voltage GaN, dc-voltage sharing, and interleaving
Uğur and Keysan [73], [74]	Multi-physics converter–machine optimization
Mohamed *et al.* [75]–[77]	Stator-mounted modules, polygon housing, integrated dc link
Wang *et al.* [78]	Repeated machine and converter submodules
Chen *et al.* [79]	WBG converter, machine, passive, and cooling co-design
Deng *et al.* [35]	Recent IMD technologies and research directions

different absolute potentials along the series dc stack. Gate-driver supplies and sensing interfaces must preserve electrical isolation between the local power stage and the control system, as illustrated by isolated traction-inverter implementations [82]. Mechanical mounting and thermal interfaces must likewise preserve the required insulation distances and solid-insulation barriers [83].

V. Modulation Strategies

Each stacked polyphase bridge (SPB) cell supports conventional sinusoidal pulsewidth modulation (SPWM), space-vector modulation (SVM), and discontinuous pulsewidth modulation (DPWM). However, the shared dc current, dynamic cell voltages, and electromagnetic coupling between winding groups require coordination across cells. Carrier phase, zero-sequence injection, switching sequence, and clamping intervals provide additional degrees of freedom for shaping dc-link ripple, machine harmonics, switching loss, common-mode voltage (CMV), and winding-group power distribution [43], [47]. Table 3 groups these degrees of freedom by their principal system-level effects.

TABLE 3. SPB modulation degrees of freedom grouped by principal system-level effect.

Effect	Degrees of freedom
Ripple and loss shaping	Carrier phase [47], [52], [84]
	Switching sequence [85], [86]
	Clamping and DPWM [87], [88]
CMV and balancing authority	Zero-sequence injection and vector redundancy [89], [90]
	Differential cell commands [55], [91]

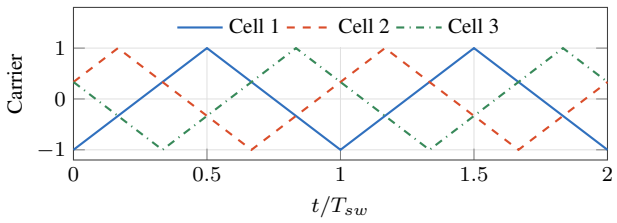


FIGURE 8. Uniform carrier interleaving for $N = 3$, with phase offsets 0, $2\pi/3$, and $4\pi/3$. Analytically generated triangular carriers.

A. Carrier Coordination and Harmonic Cancellation

For a binary upper-switch command $s_{x,k} \in \{0, 1\}$, the double-Fourier representation is

$$s_{x,k}(t) = \sum_{m,n\in\mathbb{Z}} c_{mn,x,k} e^{j[m(\omega_c t+\theta_{c,k})+n(\omega_0 t+\theta_{0,k})]}. \quad (7)$$

Here m and n are carrier and fundamental indices. The coefficients include the dc term, baseband harmonics, carrier harmonics, and sidebands; conjugate symmetry gives a real switching function. A common carrier frequency and fundamental frequency are assumed across cells.

The carrier phase is a free parameter for shaping aggregate harmonics. A general carrier assignment is

$$\theta_{c,k} = \theta_{c,1} + \delta_k, \qquad \delta_1 = 0. \quad (8)$$

Uniform interleaving uses $\delta_k = 2\pi(k-1)/N$; Fig. 8 illustrates this case for $N = 3$.

B. DC-Link Current Ripple and Battery-Side Current

DC-link current ripple affects capacitor loading and the current presented to the battery and input filter. Higher switching frequency can reduce passive-component requirements within a specified loss budget, while interleaving shapes the aggregate spectrum.

With phase current positive from converter to machine, the ideal cell input current is $i_{\mathrm{conv},k} = \sum_x s_{x,k} i_{x,k}$. Consequently, its spectrum is

$$I_{\mathrm{conv},k} = \sum_{x\in\{a,b,c\}} S_{x,k} \otimes I_{x,k}. \tag{9}$$

For matched cell spectra with a common fundamental phase, uniform interleaving multiplies carrier-order m by

$$\sum_{k=1}^{N} e^{jm2\pi(k-1)/N} = \begin{cases} N, & m \in N\mathbb{Z}, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

This selection rule cancels nonmultiples of N; it does not guarantee a nonzero component exactly at Nf_{sw}. The first surviving term also depends on the switching coefficients, sideband index, and winding/current phase relationships [43]. Unequal cell loading or voltage sharing weakens cancellation.

The shared dc-link current follows from the equivalent circuit in Fig. 5:

$$\begin{aligned} L_b \frac{di_{\mathrm{dc}}}{dt} &= E_b - R_b i_{\mathrm{dc}} - \sum_k v_{\mathrm{conv},k}, \\ C_k \frac{dv_{\mathrm{conv},k}}{dt} &= i_{\mathrm{dc}} - i_{\mathrm{conv},k}. \end{aligned} \tag{11}$$

For equal cell capacitances, $C_k = C$, the relationship between the summed converter-current spectrum and dc-bus current is [43]

$$I_{\mathrm{dc}}(\omega) = \frac{\sum_{k=1}^{N} I_{\mathrm{conv},k}(\omega)}{N - \omega^2 L_b C + j\omega C R_b}. \tag{12}$$

Equation (12) assumes identical capacitors, a fixed source voltage, and prescribed converter-current spectra. For positive-frequency peak phasors, a consistent ripple measure is

$$I_{\mathrm{dc,rms}}^{\mathrm{ripple}} = \left[\frac{1}{2} \sum_{\omega_h > 0} \frac{|\sum_k I_{\mathrm{conv},k}(\omega_h)|^2}{|N - \omega_h^2 L_b C + j\omega_h C R_b|^2} \right]^{1/2}. \tag{13}$$

Contributions at the same frequency are combined before taking the magnitude. Retained harmonics must be evaluated against the input-network resonance; interleaving does not ensure attenuation at every frequency. Fig. 9 illustrates one operating point.

The analytical example includes lumped battery-to-stack inductance but assumes identical cells, prescribed sinusoidal phase currents, ideal switching, and ideal cell connections. In distributed implementations, bus-bar and connector-loop inductances alter harmonic cancellation. For instance, a series-connected gallium nitride integrated modular motor drive required careful capacitor selection and compact layout to preserve interleaving benefits [40]; an earlier characterization found that parasitic loop inductance could make interleaved ripple exceed the noninterleaved value [92].

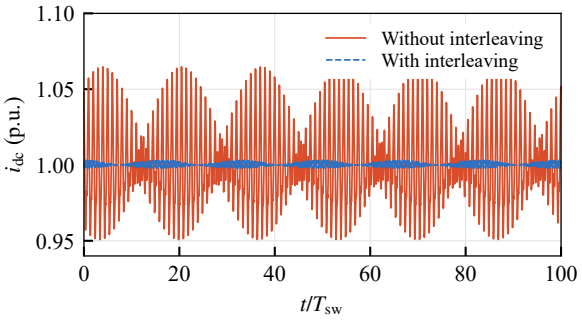


FIGURE 9. Analytical steady-state shared dc-link current for three identical SPB cells with ideal natural SPWM over one fundamental period. Parameters are $M = 0.85$, power-factor angle 20°, $f_{sw}/f_1 = 100$, $L_bC/T_{sw}^2 = 0.85$, and $R_bC/T_{sw} = 2.5$.

C. Machine Harmonics, Voltage Utilization, and Switching Loss

The preferred carrier displacement depends on cell count, modulation index, power factor, winding displacement, and the optimization objective. SPB studies demonstrate its influence on semiconductor losses and dc-link current harmonics [47], while related modular and multiphase drives demonstrate simultaneous reductions in dc-link and machine current ripple [52], [85], [93].

In multiphase machines, nonfundamental voltages can excite low-impedance harmonic subspaces, increasing copper loss and torque ripple despite satisfactory fundamental current control. Interleaved and discontinuous modulation strategies therefore require adaptation to the multiphase current-ripple characteristics [84], [94], [95].

At high speed, voltage utilization becomes critical because each winding group is supplied by its local cell voltage $V_{\mathrm{dc},k}$, subject to the machine voltage limit in (6). This constraint becomes more restrictive as cell count increases and local voltage decreases. Optimized vector selection in dual-three-phase traction machines can extend operation in the high-modulation region while suppressing unwanted subspace currents [65].

Switching-loss reduction introduces a competing objective. DPWM and hybrid strategies reduce transitions by clamping selected phases or redistributing high-frequency switching among modules. Related open-end-winding studies demonstrate simultaneous reductions in switching loss and zero-sequence current [87], [88]. Although their dc architecture differs, the principle of coordinating switching activity among inverter modules is applicable to SPB.

D. Coupling to Voltage Balancing and Common-Mode Control

Modulation also determines the control authority available for dc-link voltage balancing and CMV reduction. Voltage balancing requires differential winding-group power commands that remain feasible within local voltage and current limits [55], [56], [91]. Redundant voltage vectors and coordinated zero-sequence injection can reduce common-

mode excitation of the machine frame [89], [90], [96]. These objectives are examined in Sections VII and VI, respectively.

SPB modulation consequently requires coordinated switching decisions that account for ripple, losses, voltage utilization, balancing, and common-mode behavior across the operating map.

VI. Common-Mode Voltage Effects

In any converter-fed ac machine, the non-zero average of the three phase-leg pole voltages produced by PWM switching gives rise to a common-mode voltage (CMV) that excites parasitic capacitive paths between winding, frame, shaft, and bearings. This mechanism is well characterized for a converter with a single machine port whose voltage scaling is produced internally, such as the flying-capacitor multilevel converter, for which an analytical CM-EMI model referenced to one dc-link potential exists [97]. SPB does not fit this picture: because its N cells are connected in series on the dc side while each feeds an electrically separate winding group, cell k's local dc rails are displaced from the overall dc-bus reference by an amount that depends on its position in the stack.

A. Series-Stack Origin of Common-Mode Excitation

Taking cell 1's negative dc rail as the reference, with cell k's negative rail coinciding with cell $(k-1)$'s positive rail, the absolute offset of cell k is

$$V_{-,k} = \sum_{j=1}^{k-1} V_{\mathrm{dc},j} \approx (k-1)\frac{V_{\mathrm{dc}}}{N}, \tag{14}$$

using the voltage-partitioning relation (1). The common-mode potential seen by winding group k is then

$$v_{\mathrm{cm},k}^{\mathrm{abs}}(t) = V_{-,k} + \frac{1}{3}\left(v_{aN_k} + v_{bN_k} + v_{cN_k}\right), \tag{15}$$

where $v_{aN_k}, v_{bN_k}, v_{cN_k}$ are cell k's pole voltages referenced to its own negative rail N_k. The second term is the conventional switching-induced CMV already addressed by zero-sequence injection and vector-redundancy modulation; the first term, $V_{-,k}$, is specific to SPB, set by the cell's stack position and by the instantaneous voltage-sharing state, not by modulation alone.

Fig. 10 illustrates the resulting structure: increasing absolute potential offset with stack position, together with representative CM current paths coupling each winding group to the shared frame and chassis ground through parasitic capacitance. Because the cells and their winding groups are physically close together in the integrated implementations, these paths are short and the coupling between cells is direct rather than mediated by long cabling. Physically integrated motor-drive assemblies show that common-mode current can resonate through exactly this kind of parasitic loop, and that the resonance is not adequately predicted by a single-source CM model [96], [98]. Applying these models to SPB requires

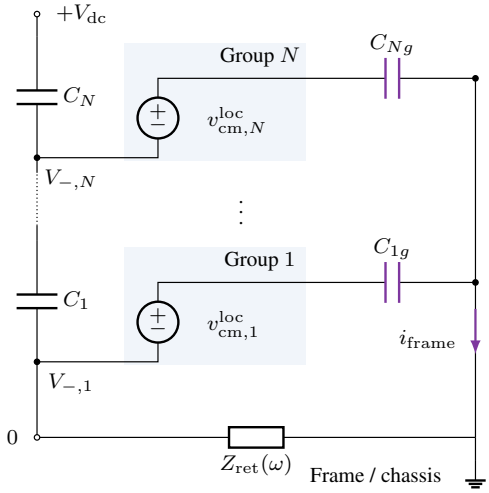


FIGURE 10. Lumped common-mode equivalent of the SPB stack.

accounting for multiple machine ports, distinct dc offsets, and their coupled return paths.

B. Carrier Synchronization for CMV Cancellation

Beyond the stack-offset mechanism above, synchronizing each cell's PWM waveform relative to the others is a way to actively cancel the switching-induced component of CMV. We illustrate the principle for the $N = 2$ case, where the cancellation condition has a simple closed form; the same synchronization idea extends to larger stacks built from matched cell pairs, subject to matching and the required winding arrangement.

For equal cell voltages V_c, define the midpoint-referenced term $u_{\mathrm{cm},k} = V_c(\sum_x s_{x,k}/3 - 1/2)$. Complementary commands $s_{x,2} = 1 - s_{x,1}$ give $u_{\mathrm{cm},2} = -u_{\mathrm{cm},1}$; reversing the second winding group's sense preserves the intended torque direction. The absolute voltages remain offset:

$$\frac{v_{\mathrm{cm},1}^{\mathrm{abs}} + v_{\mathrm{cm},2}^{\mathrm{abs}}}{2} = V_c = \frac{V_{\mathrm{dc}}}{2}. \tag{16}$$

Thus switching ripple cancels in the arithmetic mean, not the dc potential relative to the bottom rail. This balanced-cancellation principle was first demonstrated for a dual-winding-set machine driven by a balanced inverter pair on a single, non-series-stacked machine [99], and was later applied to a genuine two-cell series-stacked (2L-SSC) SPB inverter [58]; as with interleaved GaN IMMDs, the achievable reduction is ultimately limited by layout parasitics and local dc-link capacitor design [40].

This cancellation assumes perfectly matched, simultaneous switching transitions between the two cells: a propagation-delay mismatch (e.g., from PCB layout or driver differences) reintroduces a residual CM spike, of duration equal to the delay, at every misaligned transition, and unequal power sharing during dc-link balancing similarly degrades the cancellation, both shown quantitatively in [58], [99]. Fig. 11 illustrates the reference levels as an example specific to $N = 2$. For fixed lumped winding-to-frame

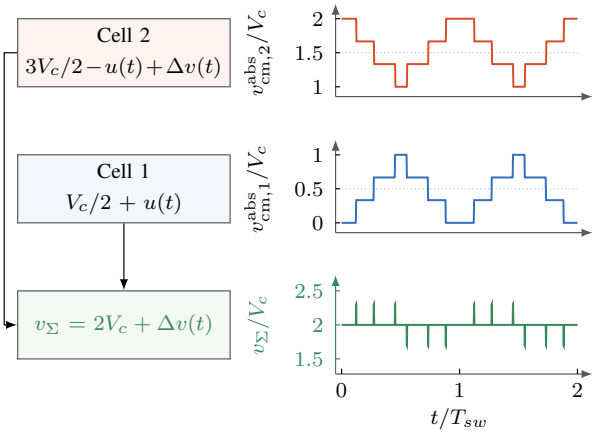


FIGURE 11. Nonideal two-cell common-mode cancellation over two switching periods. Small transition misalignment produces narrow residuals in v_Σ.

capacitances C_{kg}, the frame excitation is approximately

$$i_{\mathrm{frame}} = \sum_k C_{kg} \frac{d}{dt}(v_{\mathrm{cm},k}^{\mathrm{abs}} - v_{\mathrm{frame}}). \tag{17}$$

A constant stack offset contributes insulation stress but no steady capacitive current.

C. Physical Consequences and Mitigation

CMV couples capacitively to the rotor and can cause bearing discharges when the lubricant film breaks down [100]; non-intrusive current measurements support diagnosis [101]. In SPB, the winding groups have different absolute CM references (14)–(15). Their combined contribution to shaft voltage and bearing current, including possible resonances, requires stack-level assessment.

Fast winding-terminal transitions also stress turn-to-turn and phase-to-frame insulation. Short rise times and overshoot associated with WBG switching can accelerate insulation aging and partial discharge [102]–[104]. Insulation diagnostics [105] and suitable material systems [106] complement coordination of the local switching stress and the stack-position-dependent offset.

Candidate mitigation measures include active EMI filtering [107], shaft-voltage cancellation [108], and CMV-aware zero-sequence injection or redundant-vector selection [89], [90]. However, carrier phasing selected for ripple reduction (8) does not necessarily minimize aggregate CM excitation and cannot alter the fixed stack offset $V_{-,k}$. SPB modulation and insulation design should therefore evaluate ripple, CM excitation, and insulation stress jointly across the full stack.

VII. DC-Link Dynamics, Stability, and Voltage Balancing

The common dc current and independently controlled winding-group powers couple the local capacitor energies, so equal cell voltages are not inherently maintained [45], [46]. Related research likewise shows that distributed current control alone does not ensure stable voltage sharing [92].

For cell k, the power balance is

$$C_k V_{\mathrm{dc},k} \frac{dV_{\mathrm{dc},k}}{dt} = V_{\mathrm{dc},k} i_{\mathrm{dc}} - p_{\mathrm{ac},k} - p_{\mathrm{loss},k}. \tag{18}$$

Because the series current is common, dc input power depends on local voltage, whereas the winding-group controller determines ac output power. Defining the deviation from the mean cell voltage as

$$\widetilde{V}_k = V_{\mathrm{dc},k} - \frac{1}{N}\sum_{j=1}^{N} V_{\mathrm{dc},j}. \tag{19}$$

a simplified differential-mode model around a balanced operating point is

$$CV_0 \frac{d\widetilde{V}_k}{dt} \approx I_{\mathrm{dc},0}\widetilde{V}_k - \Delta P_k. \tag{20}$$

where C and V_0 are the common cell capacitance and equilibrium voltage, $I_{\mathrm{dc},0}$ is the equilibrium dc current, and ΔP_k is the differential output-power perturbation. During motoring, $I_{\mathrm{dc},0} > 0$, a positive voltage perturbation increases cell input power and reinforces the deviation if output power remains unchanged. During regeneration, this feedback reverses. This interpretation agrees with SPB stability analyses [45], [46] and related series-connected multiphase-drive studies [109].

The constant-power-load model considered in [46] comprises

$$\begin{aligned} L_b \frac{di_{\mathrm{dc}}}{dt} &= E_b - R_b i_{\mathrm{dc}} - \sum_k V_{\mathrm{dc},k}, \\ C_k \frac{dV_{\mathrm{dc},k}}{dt} &= i_{\mathrm{dc}} - \frac{P_k}{V_{\mathrm{dc},k}}. \end{aligned} \tag{21}$$

Here, E_b is the dc-source voltage, and L_b and R_b are the lumped series inductance and resistance of the shared source-to-stack supply path, including source and interconnection contributions; P_k is the cell's constant power demand. Under this model's assumptions, linearization about the open-loop motoring equilibrium for $N > 1$ yields $N - 1$ strictly positive real eigenvalues, establishing instability irrespective of parameter selection.

Fig. 12 illustrates the balancing principle using the small-signal model in (20), consistent with [46]. At equal constant output power, a lower-voltage cell draws more converter input current ($P_k/V_{\mathrm{dc},k}$) and its capacitor discharges further; a higher-voltage cell exhibits the opposite response. Thus, a small mismatch grows during motoring unless differential power is adjusted. The law $\Delta P_k = K_p \widetilde{V}_k$, motivated by controller alternative I in [46], increases output power in higher-voltage cells and decreases it in lower-voltage cells. For $K_p > I_{\mathrm{dc},0}$, (20) predicts exponential decay of the deviations. This local result assumes identical capacitances and ideal power-command tracking; it does not imply instability under every converter–machine control strategy.

The winding groups provide control authority over internal capacitor energy through differential power commands:

$$P_k^\star = \frac{P^\star}{N} + \Delta P_k, \qquad \sum_{k=1}^{N} \Delta P_k = 0, \tag{22}$$

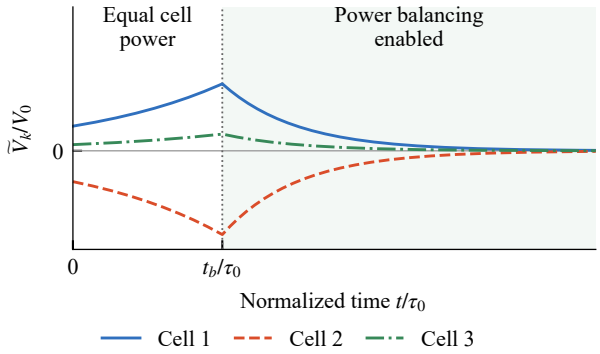


FIGURE 12. **Conceptual analytical illustration of voltage balancing for three cells during motoring, based on (20) and [46]. Small deviations from the mean grow under equal constant cell powers. Enabling differential power balancing at t_b makes them decay toward zero while $\sum_k \Delta P_k = 0$. Time is normalized by $\tau_0 = CV_0/I_{\mathrm{dc},0}$.**

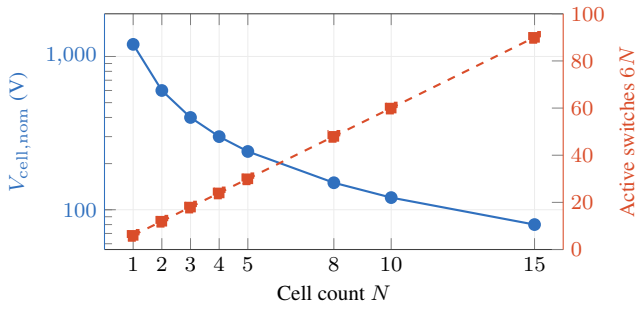
TABLE 4. **Principal SPB dc-link balancing approaches and their main considerations.**

Approach	Mechanism	Key consideration
Centralized	Differential current/power commands [45], [46]	Direct authority; global sensing and coordination
CDR	Common modulation, natural coupled dynamics [53], [56]	Low overhead; symmetry- and timing-sensitive
Distributed	Local agents, neighbor communication [54], [55], [57]	Scalable; requires communication and synchronization

This allocation redistributes cell power without changing the total machine-power reference. Available balancing authority depends on phase-current margin, winding voltage, torque-ripple limits, and thermal state.

Table 4 summarizes three principal approaches. Centralized control explicitly regulates voltage sharing through differential commands [45], [46]. Common-duty-ratio (CDR) control instead exploits the coupled converter–machine dynamics; stable sharing has been demonstrated for the investigated induction-machine system [53], with practical performance sensitive to timing mismatch [56]. Distributed multi-agent control extends local coordination to voltage balancing, power sharing, and reconfiguration [54], [55], [57]. Related modular-drive work uses local speed-droop laws to coordinate module power sharing [110].

For automotive applications, balancing must remain effective during torque transients, field weakening, parameter mismatch, and module reconfiguration. Sensor offsets, communication latency, propagation delays, and differential power limits determine the usable cell-voltage margin and therefore influence semiconductor voltage-rating selection. Dc-link stability must consequently be assessed across the coupled converter, machine, and control system.



N	1	2	3	4	5	8	10	15
V_{cell} (V)	1200	600	400	300	240	150	120	80
$6N$	6	12	18	24	30	48	60	90

FIGURE 13. **Cell-count scaling for a 1200-V dc link: nominal cell voltage $V_{\mathrm{cell,nom}} = V_{\mathrm{dc}}/N$ and active switch positions $6N$.**

VIII. Semiconductor Technology and Cell-Count Scaling

Increasing the SPB cell count N enables lower-voltage semiconductor devices, but voltage-rating selection must account for maximum blocking stress:

$$V_{\mathrm{rated}} \geq k_{\mathrm{rel}} \left(\frac{V_{\mathrm{dc,max}}}{N} + \Delta V_{\mathrm{imb}}^{\mathrm{max}} + \Delta V_{\mathrm{sw}}^{\mathrm{max}} \right), \qquad (23)$$

where $\Delta V_{\mathrm{imb}}^{\mathrm{max}}$ is the maximum cell-voltage imbalance, $\Delta V_{\mathrm{sw}}^{\mathrm{max}}$ is the commutation overshoot, and k_{rel} is the selected voltage-rating margin factor. Cell-count selection therefore depends on discrete commercial voltage classes and their usable operating margins.

SiC is well suited to high-voltage traction, while voltage partitioning can make lower-voltage GaN devices attractive through favorable conduction and switching characteristics [25], [36], [58]. For the illustrative 1200-V dc link in Fig. 13, two cells operate at a nominal 600 V each, leaving only 50 V of headroom for 650-V devices. This allocation is infeasible if bus variation, imbalance, switching overshoot, and the selected design margin require more than this headroom. Three cells reduce this voltage to 400 V, providing greater allowance for imbalance and switching transients. Further increases in N enable lower-voltage device classes but increase the number of active switch positions as $6N$. Final device selection must satisfy (23).

A. Commercial-Device Screening

Lower voltage ratings do not guarantee superior converter performance. Candidate devices must be assessed using conduction and switching characteristics together with current capability, thermal limits, and packaging constraints. Figures of merit such as $R_{\mathrm{DS(on)}}C_{\mathrm{oss}}$ and $R_{\mathrm{DS(on)}}Q_g$ provide initial screening, but detailed comparison requires $R_{\mathrm{DS(on)}}(T)$ for conduction loss and die-area requirements; $E_{\mathrm{oss}}(V)$ for stored output-capacitance energy and its contribution to switching loss; $E_{\mathrm{on/off}}(V, I, T)$ for switching-energy characterization; Q_g and Q_{gd} for gate-drive requirements and switching behavior; I_{cont} and R_θ for current derating, parallelization, and cooling; and package and power-loop inductance L_σ for commutation overshoot.

Machine phase current does not generally decrease as $1/N$. For a derated device current capability $I_{\mathrm{dev,der}}$, the required parallel-device count and total active-device count can be estimated as

$$n_p = \left\lceil \frac{I_{\mathrm{sw,max}}}{I_{\mathrm{dev,der}}} \right\rceil, \qquad N_{\mathrm{sw,tot}} = 6Nn_p. \tag{24}$$

Consequently, favorable device-level figures of merit may coexist with substantial die area, package count, and gate-drive and thermal-interface requirements.

Device selection and layout are further coupled through commutation overshoot:

$$\Delta V_{\mathrm{sw}} \approx L_\sigma \frac{di}{dt}, \tag{25}$$

Package and interconnection inductance can consume much of the voltage margin gained through partitioning. High-current GaN modules illustrate the importance of low-inductance packaging and integrated gate driving at traction-relevant currents [111], [112]. High switching speeds therefore require careful commutation-loop design, local decoupling, and gate-drive layout, particularly for GaN [112]. Dynamic $R_{\mathrm{DS(on)}}$, reverse conduction, gate reliability, short-circuit capability, and thermal performance must also be considered [113], [114]. Room-temperature static resistance alone is therefore insufficient for cell-count selection.

B. Loss Scaling With Cell Count

For an individual device, conduction and switching losses can be approximated as

$$P_{\mathrm{cond}} \approx I^2_{\mathrm{rms,sw}} R_{\mathrm{DS(on)}}(T_j), \tag{26}$$

and

$$P_{\mathrm{sw}} \approx f_{\mathrm{sw}} \left[E_{\mathrm{on}}(V, I, T_j) + E_{\mathrm{off}}(V, I, T_j) \right]. \tag{27}$$

Including dead-time-related losses and gate-drive power gives

$$P_{\mathrm{semi}}(N) = \sum_{k=1}^{N} \sum_{j=1}^{6n_p(N)} \left(P_{\mathrm{cond},k,j} + P_{\mathrm{sw},k,j} + P_{\mathrm{dt},k,j} \right) + P_{\mathrm{gate}}(N), \tag{28}$$

where $P_{\mathrm{dt},k,j}$ represents dead-time-related device loss and P_{gate} is the aggregate gate-drive power.

Total loss does not scale linearly with N: additional cells increase device count, while transitions between voltage classes change on-state resistance, switching energy, thermal impedance, and parallelization requirements. Low-cell-count designs may favor high-voltage SiC, whereas lower-voltage GaN becomes attractive when its device, packaging, or passive-component benefits offset the additional modules.

IX. Fault Tolerance and Reconfiguration

SPB modularity offers potential graceful degradation because the converter and machine are divided into repeated converter–winding modules. However, fault tolerance is not automatic: because the dc ports are connected in series, a failed cell can disturb or interrupt the complete dc current path, a structural challenge demonstrated for SPB and for other series-connected modular converters alike [115]–[117]. Early SPB work demonstrated post-fault operation under semiconductor faults, establishing that repeated hardware must be accompanied by appropriate safe states and control action [115]. At the machine level, a submodule short-circuit fault in a modular converter feeding fractional-slot concentrated-winding PM machines was shown to drive a fault current above rated current because of the strong NdFeB rotor magnetization, risking machine overheating; reducing rotor magnetization limits this fault current but also narrows the constant-power-speed range, a design tradeoff specific to fault-tolerant FSCW PM machines [116]. In the SPB machine designs of that study, post-fault short-circuit current with NdFeB magnets exceeds the healthy-phase current, whereas with ferrite magnets it stays below it, at the cost of becoming visibly non-sinusoidal from magnetic coupling with the adjacent healthy windings – a concrete illustration that PM material selection is itself a fault-tolerance design variable, not only a torque-density one. Fault handling in other series-connected modular converters further underscores that mitigation requires the control system to be explicitly designed for module isolation [117].

The machine provides a second layer of redundancy. Healthy winding groups can potentially redistribute current after phase or converter faults, as demonstrated in recent multiphase and modular-drive studies [63], [118], [119]. In SPB, however, this machine-side redundancy is useful only if the failed converter cell can be isolated or bypassed without breaking the series dc path.

Building on the reported online reconfiguration capability [57], after a faulted module is isolated the remaining active set $\mathcal{A}$ must satisfy

$$P^\star = \sum_{k \in \mathcal{A}} P^\star_k, \tag{29}$$

with torque derating if current, voltage, or thermal limits prevent full redistribution. Reconfiguration can therefore provide both fault isolation and an operating-range degree of freedom because the number of active cells changes the voltage available to the remaining winding groups. Bypassing a cell raises the steady voltage of each surviving cell when the total bus voltage remains fixed. With $M = |\mathcal{A}|$ active cells, a necessary blocking-voltage condition is

$$\frac{V_{\mathrm{dc,max}}}{M} + \Delta V_{\mathrm{imb}} + \Delta V_{\mathrm{sw}} \le V_{\mathrm{allow}}, \tag{30}$$

where V_{allow} is the allowable device voltage after the chosen design margin, and the two increments cover cell imbalance and switching overshoot. Capacitor and insulation limits must also hold. Current or torque derating cannot by itself correct excessive steady blocking voltage. If this condition fails, operation requires a lower bus voltage, a suitable spare-cell arrangement, or shutdown. Fig. 15 separates voltage feasibility from post-fault torque capability. Fig. 14 maps these fault locations onto the SPB structure.

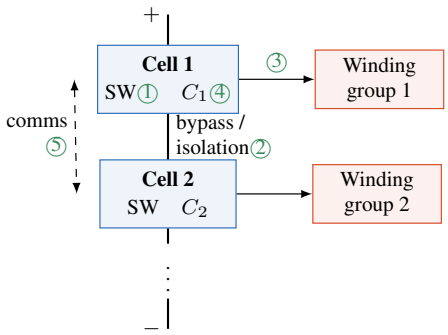


FIGURE 14. Simplified SPB converter–machine structure with fault locations ①–⑤ annotated, corresponding in order to the rows of Table 5.

TABLE 5. SPB Fault Locations and Required System Response

Fault	SPB consequence	Required response
Switch / bridge leg	Cell cannot synthesize commanded voltage	Fast safe state and protection [115]
Converter module	Series dc path can be interrupted	Bypass/isolation and power redistribution [57]
Winding / phase	Reduced and asymmetric torque capability	Multiphase post-fault current control [63], [119]
Capacitor / sensor*	Loss of voltage-sharing information or margin	Detection, derating, isolation or shutdown
Communication / controller	Loss of distributed coordination	Local fallback and deterministic safe state [120]

*The SPB studies surveyed here do not establish capacitor or sensor fault coverage; this row reflects the authors' system-level inference.

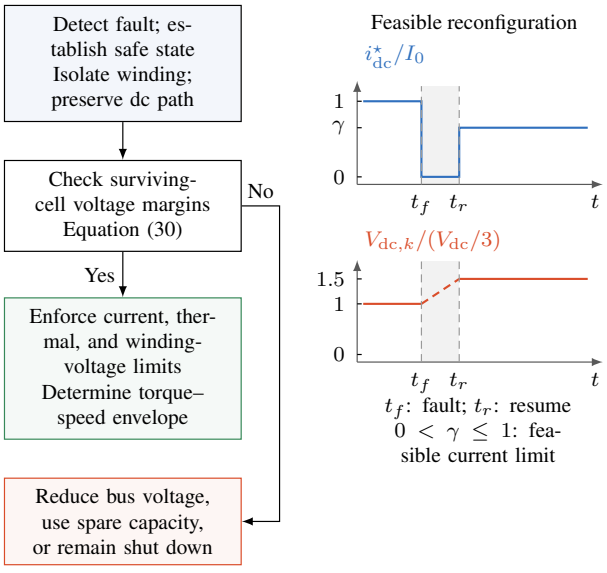


FIGURE 15. Post-fault feasibility assessment with illustrative waveforms for the feasible branch. The dc-current command is temporarily set to zero and restored to γI_0, subject to operating limits. For a fixed bus, changing from three to two active cells raises the balanced surviving-cell voltage from 1 to 1.5 p.u. (400 to 600 V for a 1200-V bus).

TABLE 6. 300-kW SPB design inputs and preliminary repeated-block loss estimate

Output power / dc-bus voltage	300 kW / 1200 V
Cells / nominal cell voltage	16 / 75 V
Devices per switch / drivers per leg	4 / 2
Total devices / half-bridge drivers	384 / 96
Phase current / switching frequency	205.0 A rms / 50 kHz
MLCCs per phase leg	$48 \times 10 + 24 \times 1$ μF
Cell capacitance, nominal / effective	1512 / 464.4 μF
PCB footprint per phase leg	16.92 cm^2
Estimated efficiency / stage loss	99.60% / 1.20 kW
Worst evaluated dc-link ripple	4.12% (target 4%)
PCB-area power density	0.37 kW/cm^2
Original two-parallel block: extracted layout	
Board size / layer count	30 × 28.2 mm / 4
Commutation loop, including capacitor ESL	0.374 nH/local loop
Gate loop / Kelvin common-source inductance	1.7 / 0.030 nH

Loss component	Leg (W)	Stack (W)
Conduction	11.66	559.5
Switching	3.87	185.9
PCB copper (load)	7.44	356.9
PCB copper (harmonics)	0.99	47.7
Capacitor ESR	0.45	21.4
Dead time	0.41	19.8
Gate drive and LDO	0.16	7.5
Total	24.97	1198.7

Reliability remains ambiguous because increasing cell count provides more potential degraded states but also more components that can fail. System or mission reliability is therefore more meaningful than component count alone. Markov-chain analysis has been applied to related fault-tolerant IMMDs to distinguish healthy, degraded, and failed states [121], while integrated distributed traction drives provide additional evidence that fault tolerance must include both the machine and the local converter architecture [71], [72].

The SPB literature therefore demonstrates the feasibility of post-fault operation and online reconfiguration, but not yet automotive-level fault coverage. Detection and isolation time, short-circuit energy, capacitor and sensor faults, communication loss, post-fault thermal loading, diagnostic coverage, and deterministic functional-safety transitions remain important validation topics.

X. Illustrative Device-to-System Design Example

Fig. 16 synthesizes the preceding review into an iterative device-to-system design procedure. Electrical specifications and the operating mission define the constraints; device voltage class, cell count, layout, parallel count, capacitance, and switching frequency are then evaluated together with

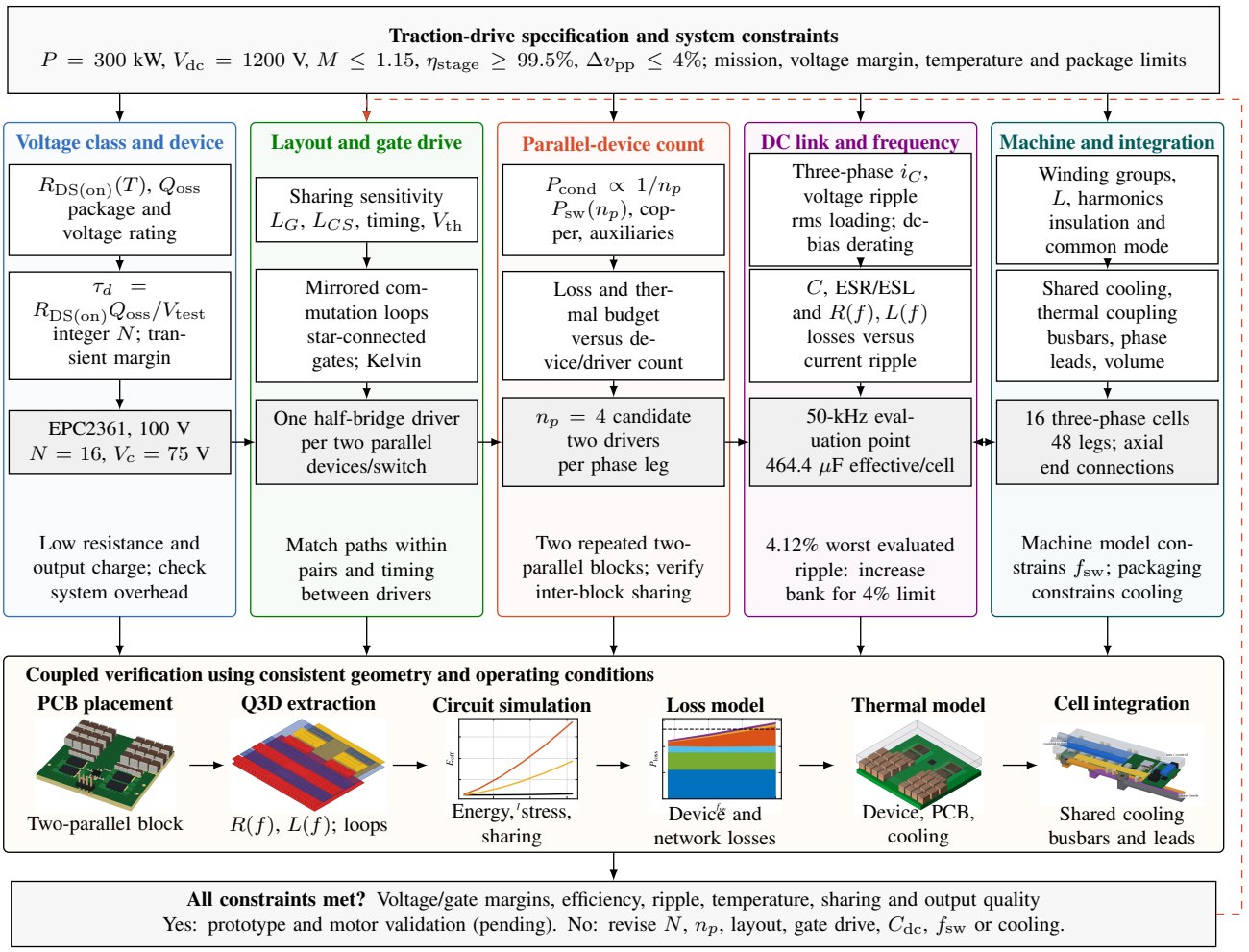


FIGURE 16. **Iterative SPB system-design procedure with coupled verification and cell integration. Machine requirements feed back into switching-frequency, insulation, and thermal design. Original two-parallel building-block models support the four-parallel analytical projection; shared cooling and interconnections require assembly-level and experimental validation.**

the machine and cooling arrangement. The verification loop revisits these choices until electrical, thermal, and packaging requirements are met. It provides a framework for system optimization, rather than demonstrating a global optimum.

The example in Table 6 uses a 300-kW, 1200-V SPB. For balanced sinusoidal operation, $I_{ph,rms} = 2\sqrt{2}P_{out}/(3MV_{dc}\cos\varphi)$; $M = 1.15$ and unity power factor give 205.0 A rms. The EPC2361 is selected as an illustrative 100-V candidate [122]. Sixteen cells operate at 75 V nominal; maximum-bus operation, balancing error, and overshoot must still satisfy (23).

Four devices per switch are represented by four-parallel blocks with matched Kelvin gate paths and two half-bridge drivers per leg. At 50 kHz, the loss model calculates conduction losses scaled, sinusoidally averaged switching-energy fits, dead-time reverse conduction, frequency-dependent PCB resistance, capacitor ESR, and gate-drive losses. It assumes equal block currents and unchanged local parasitics. The three-phase capacitor network accounts for dc-bias capacitance reduction and ESR/ESL-dependent current sharing; the listed bank requires a modest increase to meet the 4% ripple target over the evaluated operating points.

The projected power-stage efficiency is 99.60%, excluding external isolated supplies, control electronics, inter-block busbars, cooling auxiliaries, and machine losses. PCB-area density excludes the cooling assembly and enclosure. Extracted inductances describe the original block, with package and unmodeled routing inductance excluded from the commutation value.

XI. Research Gaps and Future Directions

SPB competitiveness depends on joint battery, converter, and machine design. Voltage partitioning, winding constraints, modulation, balancing, common-mode voltage, and fault tolerance must be evaluated together.

The optimum cell count N remains the central unresolved design question: the voltage-versus-component-count tradeoff shifts the optimum as SiC, GaN, packaging, and cooling evolve [3], [25]. A common-platform comparison of a few-cell 650-V GaN SPB against a many-cell low-

voltage GaN SPB, using real commercial devices, would be valuable; the GaN comparison in [58] is a useful starting point but should be extended to automotive power with current parallelization, machine voltage/speed constraints, dc-link capacitance, cooling, and mission energy.

Converter, machine, and thermal models, among others, should be co-optimized rather than designed sequentially. Existing phase-count studies show machine design changes with SPB cell number [34]; the missing step is a complete automotive co-design using vehicle-level energy, volume, mass, cost, and reliability as a joint objective.

Reliability is part of that objective: future studies should combine semiconductor, capacitor, sensor, gate-driver, communication, and winding faults [57], [115] in one state-based, Markov-style framework as used for related fault-tolerant IMMDs [121], keeping *component failure*, *module isolation*, and *loss of vehicle function* distinct. An N-cell drive may survive with $N-1$ active modules, but its torque–speed envelope, module thermal loading, and dc-voltage margin all change, so reliability data should pair with a post-fault capability model – not a series reliability block diagram – to compare availability, limp-home power, and mission completion probability against efficiency.

Cost and sustainability assessment should use the same N-dependent framing: increasing N can reduce per-switch die cost but tends to increase isolated gate supplies, sensing channels, and assembly operations, among others, almost linearly, while a repeated converter–winding module can gain from common tooling and automated assembly – a balance likely to differ between high-volume passenger vehicles and lower-volume heavy-duty systems. The same accounting should separate production, operation, maintenance, and end-of-life impact over one vehicle mission and service life, since modular repair and longer service life can offset the added production impact of extra components.

Overall, SPB research must move from demonstrating that the architecture operates toward demonstrating where it is better: whether it gives a measurable vehicle-level advantage over mature two- and three-level architectures once machine design, thermal management, EMC, reliability, manufacturing, and cost are included.

XII. Conclusion

This paper reviewed SPB converters – series-stacked two-level bridges each feeding an electrically separated winding group – as a candidate architecture for future high-voltage traction drives, partitioning the dc-link voltage while introducing converter–machine modularity, distributed control, and fault-management opportunities.

The review also shows that the semiconductor voltage advantage cannot be separated from the complete traction system. Increasing cell count adds converter, control, and insulation overhead, so SPB does not eliminate the complexity of high-voltage conversion but redistributes it between semiconductor technology, machine architecture, and control. WBG development makes this trade increasingly relevant, particularly where SPB can move a high-voltage traction system toward lower-voltage GaN devices; whether that approach is advantageous depends on whether the device-level gains compensate for the additional modular hardware and machine requirements.

The remaining priority is integrated co-design across the battery/dc-link, semiconductor/converter, and machine sides, evaluated jointly against vehicle-level energy, cost, and reliability targets rather than compared through separate, narrower studies. The proposed design framework makes the evaluation boundary explicit. SPB is a modular traction-drive architecture that exchanges semiconductor blocking-voltage stress for converter–machine modularity. Its competitiveness depends on whether that exchange produces a measurable vehicle-level benefit.

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REFERENCES

[1] Y. Vijaya Sambhavi and V. Ramachandran, "A technical review of modern traction inverter systems used in electric vehicle application," *Energy Reports*, vol. 10, pp. 3882–3907, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352484723014889

[2] J. Kwak and A. Castellazzi, "State-of-the-art 800 v electric drive systems: Inverter–machine codesign for energy efficiency optimization," *Electronics*, vol. 12, no. 14, 2023. [Online]. Available: https://www.mdpi.com/2079-9292/12/14/3063

[3] M. I. I. Sakib, S. Jaman, B. Verbrugge, M. E. Baghdadi, S. Chakraborty, and O. Hegazy, "Electro-thermal, emi and reliability assessment of post-800 v traction inverter topologies," *World Electr. Veh. J.*, vol. 17, no. 8, 2026. [Online]. Available: https://www.mdpi.com/2032-6653/17/8/384

[4] I. Aghabali, J. Bauman, P. J. Kollmeyer, Y. Wang, B. Bilgin, and A. Emadi, "800-v electric vehicle powertrains: Review and analysis of benefits, challenges, and future trends," *IEEE Trans. on Transp. Electrification*, vol. 7, no. 3, pp. 927–948, 2021.

[5] M. H. Mousavi, H. Moradi, and K. Rouzbehi, "Transition of dc link voltage from 400v to 800v in electric vehicles: Performance, trade-offs, and technical insights," in *2025 10th International Conference on Technology and Energy Management (ICTEM)*, 2025, pp. 1–7.

[6] T. Bertelshofer, M. Denk, and M.-M. Bakran, "Design study and prototype of 150 kw inverter with discrete sic mosfets," in *PCIM Europe 2019; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management*, 2019, pp. 1–8.

[7] Bosch Mobility, "Inverter generation 4," product specification: 400-V version up to 250 kW peak; 800-V version up to 350 kW peak. Accessed Aug. 28, 2026.

[8] ADAC, "Bmw i4 m50 technical data," 2025, manufacturer data: 400-V high-voltage architecture and 400-kW maximum system power. Accessed Aug. 28, 2026.

[9] W. Taha, F. Juarez-Leon, M. Hefny, A. Jinesh, M. Poulton, B. Bilgin, and A. Emadi, "Holistic design and development of a 100-kw sic-based six-phase traction inverter for an electric vehicle application," *IEEE Trans. on Transp. Electrification*, vol. 10, no. 2, pp. 4616–4627, 2024.

[10] Texas Instruments and Wolfspeed, "TIDM-02014: High-power, high-performance automotive SiC traction inverter reference design," 2026, 800-V, 300-kW SiC traction-inverter system reference design. Accessed: Aug. 28, 2026. [Online]. Available: https://www.ti.com/tool/TIDM-02014

[11] Hyundai Motor Company, "IONIQ 5 N performance and technical specifications," 2026, 800-V battery architecture and up to 478-kW output. Accessed: Aug. 28, 2026. [Online]. Available: https://www.hyundai-n.com/en/models/n/ioniq-5-n

[12] A. Al-Hmoud, A. Ismail, and Y. Zhao, "A high-density 200-kw all silicon carbide three-phase inverter for traction applications," in *2023 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2023, pp. 3143–3146.

[13] Infineon Technologies, "REF-CAV250KMT7INV: 250-kw three-phase inverter power block for 800-v eCAV traction systems," 2026, reference design. Accessed: Aug. 28, 2026. [Online]. Available: https://www.infineon.com/cms/en/product/evaluation-boards/ref-cav250kmt7inv

[14] Volvo Cars, "EX90 technical specifications," 2026, 800-V electrical system and 500-kW maximum system output for the Twin Motor Performance version. Accessed: Aug. 28, 2026. [Online]. Available: https://www.volvocars.com/us/cars/ex90-electric/specifications/

[15] J. Wang, I. Laird, X. Yuan, and W. Zhou, "A 1.2 kv 100 kw four-level anpc inverter with sic power modules and capacitor voltage balance for ev traction applications," in *2021 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2021, pp. 5202–5209.

[16] AVL Software and Functions GmbH, "PI850e 800-v SiC traction inverter platform," 2026, system voltage 800 V; power up to 250 kW. Accessed: Aug. 28, 2026. [Online]. Available: https://www.avl-functions.com/en/800v-sic-traction-inverter/

[17] Porsche AG, "Taycan turbo S technical specifications," 2026, 800-V traction architecture and up to 700-kW overboost output with Launch Control. Accessed: Aug. 28, 2026. [Online]. Available: https://www.porsche.com/international/models/taycan/taycan-models/taycan-turbo-s/

[18] E. Ayaz, S. S. Ghahfarokhi, S. Norrga, and H.-P. Nee, "Dc-link voltage scaling to 1.2 kv for heavy-duty evs: System-level assessment with experimental validation of sic traction inverter," *IEEE Access*, vol. 14, pp. 80 839–80 852, 2026.

[19] Punch Powertrain, "IV5 800-v SiC high-power inverter," 2026, product specification: up to 600 kW. Accessed: Aug. 28, 2026. [Online]. Available: https://punchpowertrain.com/products/inverters/

[20] Lucid Group, "Lucid air dream edition performance: Pricing and specifications," 2022, 900-V-plus electrical architecture and 828-kW combined motor output. Accessed: Aug. 28, 2026. [Online]. Available: https://ir.lucidmotors.com/node/7801/pdf

[21] J. Millán, P. Godignon, X. Perpiñà, A. Pérez-Tomás, and J. Rebollo, "A survey of wide bandgap power semiconductor devices," *IEEE Trans. on Power Electron.*, vol. 29, no. 5, pp. 2155–2163, 2014.

[22] F. Udrea, G. Deboy, and T. Fujihira, "Superjunction power devices, history, development, and future prospects," *IEEE Trans. on Electron Devices*, vol. 64, no. 3, pp. 713–727, 2017.

[23] J. Azurza Anderson, G. Zulauf, J. W. Kolar, and G. Deboy, "New figure-of-merit combining semiconductor and multi-level converter properties," *IEEE Open J. Power Electron.*, vol. 1, pp. 322–338, 2020.

[24] A. S. Abdelrahman, Z. Erdem, Y. Attia, and M. Z. Youssef, "Wide bandgap devices in electric vehicle converters: A performance survey," *Can. J. Electr. Comput. Eng.*, vol. 41, no. 1, pp. 45–54, 2018.

[25] H. A. Soomro, M. H. B. M. Khir, S. A. B. M. Zulkifli, G. E. M. Abro, and M. M. Abualnaeem, "Applications of wide bandgap semiconductors in electric traction drives: Current trends and future perspectives," *Results Eng.*, vol. 26, p. 104679, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S259012302500756X

[26] D. Nehmer, M. Hepp, W. Wondrak, and M.-M. Bakran, "A comparison of gan hemt and sic mosfet power inverter modules for electric vehicles (ev)," in *PCIM Europe 2023; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management*, 2023, pp. 1–9.

[27] A. Poorfakhraei, M. Narimani, and A. Emadi, "A review of multilevel inverter topologies in electric vehicles: Current status and future trends," *IEEE Open J. Power Electron.*, vol. 2, pp. 155–170, 2021.

[28] Y. Wang, R. Ilka, J. Camp, and J. He, "An immersion cooled sic t-type traction inverter–thermal performance evaluation," *IEEE Trans. on Power Electron.*, vol. 41, no. 2, pp. 1621–1626, 2026.

[29] L. Horowitz, J. Zou, and R. C. N. Pilawa-Podgurski, "Design and optimization of a high-performance 3d-stacked flying capacitor multilevel inverter for electric drivetrains," in *2025 IEEE 26th Workshop on Control and Modeling for Power Electronics (COMPEL)*, 2025, pp. 1–6.

[30] F. Gemma, J. Riccio, G. Tresca, A. Volpini, and P. Zanchetta, "Finite-control-set model predictive control with reduced computational burden in cascaded h-bridge permanent magnet synchronous motor drives for ev applications," in *2024 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2024, pp. 3941–3947.

[31] G. Choi, Z. Xu, M. Li, S. Gupta *et al.*, "Development of integrated modular motor drive for traction applications," *SAE Int. J. Engines*, vol. 4, no. 1, pp. 286–300, 2011.

[32] J. Wang, T. M. Jahns, P. McCluskey, J. Kizito, B. Sarlioglu, R. Borjas, J. A. Swanke, Y. Cong, Z. Yao, H. Zeng, P. Fu, and A. Schnabel, "2-kv 1-mw 20000-r/min integrated modular motor drive for electrified aircraft propulsion," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 13, no. 1, pp. 394–407, 2025.

[33] T. M. Jahns and H. Dai, "The past, present, and future of power electronics integration technology in motor drives," *CPSS Trans. on Power Electron. Appl.*, vol. 2, no. 3, pp. 197–216, 2017.

[34] T. Bringezu and J. Biela, "Comparison of optimized motor-inverter systems using a stacked polyphase bridge converter combined with a 3-, 6-, 9-, or 12-phase pmsm," in *2020 22nd European Conference on Power Electronics and Applications (EPE'20 ECCE Europe)*, 2020, pp. P.1–P.11.

[35] X. Deng, B. Mecrow, T. M. Jahns, L. Empringham, L. d. Lillo, N. Simpson, and A. Hopkins, "Integrated motor drives: A review of recent developments, enabling technologies, and future research directions," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 14, no. 4, pp. 4151–4171, 2026.

[36] A. Ebrahimian, S. Iman Hosseini Sabzevari, W. A. Khan, and N. Weise, "Design and experimental validation of a high-power-density gan-based inverter module for integrated modular motor drives in all-electric aircraft applications," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 13, no. 4, pp. 4504–4520, 2025.

[37] S. Norrga, L. Jin, O. Wallmark, A. Mayer, and K. Ilves, "A novel inverter topology for compact ev and hev drive systems," in *IECON 2013 - 39th Annual Conference of the IEEE Industrial Electronics Society*, 2013, pp. 6590–6595.

[38] E. Cengelci, P. Enjeti, and J. Gray, "A new modular motor-modular inverter concept for medium-voltage adjustable-speed-drive systems," *IEEE Trans. on Ind. Appl.*, vol. 36, no. 3, pp. 786–796, 2000.

[39] J. Wang, Y. Li, and Y. Han, "Evaluation and design for an integrated modular motor drive (immd) with gan devices," in *2013 IEEE Energy Conversion Congress and Exposition*, 2013, pp. 4318–4325.

[40] L. Wang and Y. Han, "Integrated modular motor drive design with gan power fets," *IEEE Trans. on Ind. Appl.*, vol. 51, no. 4, pp. 3198–3207, 2015.

[41] J. Wang and Y. Han, "A class of modular multilevel motor drives with design flexibility and reduced components number," in *2015 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2015, pp. 2387–2393.

[42] A. Ruseler, T. B. Lazzarin, and I. Barbi, "A novel modular multilevel inverter for medium-voltage open-end split winding machines based on single-phase submodules with input-series connection," in *2015 IEEE 13th Brazilian Power Electronics Conference and 1st Southern Power Electronics Conference (COBEP/SPEC)*, 2015, pp. 1–6.

[43] L. Jin, S. Norrga, O. Wallmark, and M. N. Harnefors, "Control and modulation of the stacked polyphase bridges inverter," in *2014 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2014, pp. 3023–3029.

[44] H. S. Che, E. Levi, M. Jones, M. J. Duran, W.-P. Hew, and N. A. Rahim, "Operation of a six-phase induction machine using series-connected machine-side converters," *IEEE Trans. on Ind. Electron.*, vol. 61, no. 1, pp. 164–176, 2014.

[45] M. N. Harnefors, L. Jin, L. Harnefors, O. Wallmark, M. Leksell, and S. Norrga, "Analysis of the dc-link stability for the stacked polyphase bridges converter," in *2015 17th European Conference on Power Electronics and Applications (EPE'15 ECCE-Europe)*, 2015, pp. 1–6.

[46] M. Nikouie, O. Wallmark, L. Jin, L. Harnefors, and H.-P. Nee, "Dc-link stability analysis and controller design for the stacked polyphase bridges converter," *IEEE Trans. on Power Electron.*, vol. 32, no. 2, pp. 1666–1674, 2017.

[47] L. Jin, S. Norrga, O. Wallmark, and N. Apostolopoulos, "Modulation and power losses of a stacked polyphase bridge converter," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 5, no. 1, pp. 409–418, 2017.

[48] L. Jin, S. Norrga, O. Wallmark, and N. Apostolopoulos, "Communication-based distributed control of the stacked polyphase bridges converter," *IEEE Trans. on Ind. Electron.*, vol. 65, no. 2, pp. 1011–1020, 2018.

[49] H. Zhang, L. Jin, O. Wallmark, and S. Norrga, "Evaluation of modular integrated electric drive concepts for automotive traction applications," *IEEE Trans. on Transp. Electrification*, 2017, preprint, KTH DiVA `urn:nbn:se:kth:diva-207762`; published-version DOI not located. [Online]. Available: http://kth.diva-portal.org/smash/get/diva2:1098046/FULLTEXT01.pdf

[50] L. Verkroost, J. Van Damme, H. Vansompel, F. De Belie, and P. Sergeant, "Module connection topologies and interleaving strategies for integrated modular motor drives," in *2019 IEEE International Electric Machines & Drives Conference (IEMDC)*, 2019, pp. 559–564.

[51] J. Van Damme, L. Verkroost, H. Vansompel, F. De Belie, and P. Sergeant, "A holistic dc link architecture design method for multi-phase integrated modular motor drives," in *2019 IEEE International Electric Machines & Drives Conference (IEMDC)*, 2019, pp. 1593–1598.

[52] L. Verkroost, J. Van Damme, D. V. Bozalakov, F. De Belie, P. Sergeant, and H. Vansompel, "Simultaneous dc-link and stator current ripple reduction with interleaved carriers in multiphase controlled integrated modular motor drives," *IEEE Trans. on Ind. Electron.*, vol. 68, no. 7, pp. 5616–5625, 2021.

[53] S. Mocevic, E. Aeloiza, D. Tremelling, P. Kadachigov, and L. Harnefors, "Dc-link stability of stacked polyphase bridges supplying multi-phase induction machine having common-duty-ratio control," in *2023 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2023, pp. 4652–4659.

[54] L. Verkroost, A. Vande Ghinste, K. Vanthuyne, J. Van Damme, F. De Belie, P. Sergeant, and H. Vansompel, "Multi-agent voltage balancing in modular motor drives with series-connected power electronic converters," *IET Power Electron.*, vol. 17, no. 9, pp. 1067–1076, 2024. [Online]. Available: https://ietresearch.onlinelibrary.wiley.com/doi/abs/10.1049/pel2.12452

[55] L. Verkroost, F. De Belie, L. F. da Rocha, P. K. Olsen, P. Sergeant, and H. Vansompel, "Multi-agent control scheme for power distribution in a multi-three-phase motor drive fed by a stacked polyphase bridges converter," *IEEE Trans. on Energy Convers.*, vol. 39, no. 4, pp. 2568–2580, 2024.

[56] S. Mocevic, E. Aeloiza, and P. Cairoli, "Stacked polyphase bridges converter having common-duty-ratio control with gate signal delay," in *2024 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2024, pp. 6181–6187.

[57] L. Verkroost, A. Vande Ghinste, L. F. da Rocha, J. D. M. De Kooning, F. De Belie, P. Sergeant, P. K. Olsen, and H. Vansompel, "Dynamic multi-agent dc-bus reconfiguration in modular motor drives with a stacked polyphase bridge converter," *IET Electr. Power Appl.*, vol. 18, no. 2, pp. 160–173, 2024. [Online]. Available: https://ietresearch.onlinelibrary.wiley.com/doi/abs/10.1049/elp2.12376

[58] G. Rohner, J. Huber, S. Mirić, and J. W. Kolar, "Comparative evaluation of three-phase three-level flying capacitor and stacked polyphase bridge gan inverter systems for integrated motor drives," *Electronics*, vol. 13, no. 7, 2024. [Online]. Available: https://www.mdpi.com/2079-9292/13/7/1259

[59] H. Zhang, O. Wallmark, M. Leksell, S. Norrga, M. Nikouei Harnefors, and L. Jin, "Machine design considerations for an mhf/spb-converter based electric drive," in *IECON 2014 - 40th Annual Conference of the IEEE Industrial Electronics Society*, 2014, pp. 3849–3854.

[60] M. Zhang, M. Yuan, and J. Jiang, "A comprehensive review of the multiphase motor drive topologies for high-power electric vehicle: Current status, research challenges, and future trends," *IEEE Trans. on Transp. Electrification*, vol. 11, no. 1, pp. 3631–3654, 2025.

[61] S. K. Das, A. Shukla, and S. Basak, "Comparative analysis of winding configurations for six-phase interior permanent magnet synchronous machines in electric vehicle," in *2025 IEEE 4th Industrial Electronics Society Annual On-Line Conference (ONCON)*, 2025, pp. 1–6.

[62] S. Wang, H. Fan, and J. Zhao, "Model-free predictive current control of dtp-pmsm based on ultra-local model considering back emf harmonics," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 13, no. 1, pp. 553–563, 2025.

[63] H. Tang, Z. Dong, P. Luo, C. He, G. Han, and H. Cheng, "Fault-tolerant control for the asymmetric six-phase permanent magnet synchronous motor with open-phase faults," *IEEE Trans. on Transp. Electrification*, vol. 11, no. 2, pp. 5793–5808, 2025.

[64] Y. Han, "Design, modeling, and control of multilevel converter motor drive with modular design and split winding machine," in *2014 IEEE 15th Workshop on Control and Modeling for Power Electronics (COMPEL)*, 2014, pp. 1–10.

[65] D. Chen, T. Zou, M. Cui, A. Trentin, D. Prystupa, J. Riccio, M. Degano, C. Gerada, and K. Paciura, "Current harmonic suppression for extended high-modulation operation of 250 kw high-speed dual three-phase ipm traction motor," *IEEE Trans. on Transp. Electrification*, pp. 1–1, 2026.

[66] W. Feng, F. Chen, S. Lee, K. Chen, J. Paddock, T. M. Jahns, and B. Sarlioglu, "Design and comparison of power-dense surface inset permanent magnet machine and surface permanent magnet machine without heavy rare earth magnets for traction applications," *IEEE Trans. on Ind. Appl.*, vol. 62, no. 4, pp. 6464–6477, 2026.

[67] Y. Shakweh, G. Owen, D. Hall, and H. Miller, "Plug and play integrated motor drives," in *2002 International Conference on Power Electronics, Machines and Drives (Conf. Publ. No. 487)*, 2002, pp. 655–661.

[68] P. Wheeler, J. Clare, M. Apap, L. Empringham, K. Bradley, S. Pickering, and D. Lampard, "A fully integrated 30 kw motor drive using matrix converter technology," in *2005 European Conference on Power Electronics and Applications*, 2005, pp. 9 pp.–P.9.

[69] S. Pickering, P. Wheeler, F. Thovex, and K. Bradley, "Thermal design of an integrated motor drive," in *IECON 2006 - 32nd Annual Conference on IEEE Industrial Electronics*, 2006, pp. 4794–4799.

[70] N. R. Brown, T. M. Jahns, and R. D. Lorenz, "Power converter design for an integrated modular motor drive," in *2007 IEEE Industry Applications Annual Meeting*, 2007, pp. 1322–1328.

[71] J. Wolmarans, M. Gerber, H. Polinder, S. de Haan, J. Ferreira, and D. Clarenbach, "A 50kw integrated fault tolerant permanent magnet machine and motor drive," in *2008 IEEE Power Electronics Specialists Conference*, 2008, pp. 345–351.

[72] M. D. Hennen, M. Niessen, C. Heyers, H. J. Brauer, and R. W. De Doncker, "Development and control of an integrated and distributed inverter for a fault tolerant five-phase switched reluctance traction drive," *IEEE Trans. on Power Electron.*, vol. 27, no. 2, pp. 547–554, 2012.

[73] M. Ugur and O. Keysan, "Design of a gan based integrated modular motor drive," in *2018 XIII International Conference on Electrical Machines (ICEM)*, 2018, pp. 1471–1477.

[74] M. Ugur and O. Keysan, "Multi-physics design optimisation of a gan-based integrated modular motor drive system," *The J. Eng.*, vol. 2019, no. 17, 2019.

[75] A. H. Mohamed, H. Vansompel, and P. Sergeant, "Electrothermal design of a discrete gan-based converter for integrated modular motor drives," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 9, no. 5, pp. 5390–5406, 2021.

[76] A. H. Mohamed, H. Vansompel, and P. Sergeant, "An integrated motor drive with enhanced power density using modular converter structure," in *2021 IEEE International Electric Machines & Drives Conference (IEMDC)*, 2021, pp. 1–6.

[77] A. H. Mohamed, H. Vansompel, and P. Sergeant, "Design of an integrated dc-link structure for reconfigurable integrated modular motor drives," *IEEE Trans. on Ind. Electron.*, vol. 69, no. 3, pp. 2312–2321, 2022.

[78] J. Wang, T. M. Jahns, P. McCluskey, J. Kizito, B. Sarlioglu, R. Borjas, J. A. Swanke, Y. Cong, Z. Yao, H. Zeng, P. Fu, and A. Schnabel, "2-kv 1-mw 20000-r/min integrated modular motor drive for electrified aircraft propulsion," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 13, no. 1, pp. 394–407, 2025.

[79] F. Chen, S. Lee, W. Feng, K. Chen, J. Paddock, T. M. Jahns, and B. Sarlioglu, "Power-dense integrated motor drive using a wbg-

enabled current-source inverter for ev traction applications," *IEEE Trans. on Transp. Electrification*, vol. 12, no. 1, pp. 542–555, 2026.
[80] S. Chowdhury, E. Gurpinar, G.-J. Su, T. Raminosoa, T. A. Burress, and B. Ozpineci, "Enabling technologies for compact integrated electric drives for automotive traction applications," in *2019 IEEE Transportation Electrification Conference and Expo (ITEC)*, 2019, pp. 1–8.
[81] R. Abebe, G. Vakil, G. Lo Calzo, T. Cox, S. Lambert, M. Johnson, C. Gerada, and B. Mecrow, "Integrated motor drives: state of the art and future trends," *IET Electr. Power Appl.*, vol. 10, no. 8, pp. 757–771, 2016. [Online]. Available: https://ietresearch.onlinelibrary.wiley.com/doi/abs/10.1049/iet-epa.2015.0506
[82] Texas Instruments, "SiC/IGBT isolated gate driver reference design with thermal diode and sensing FET," Texas Instruments, Tech. Rep. TIDUEV9, Dec. 2019. [Online]. Available: https://www.ti.com/lit/ug/tiduev9/tiduev9.pdf
[83] *Insulation Coordination for Equipment Within Low-Voltage Supply Systems—Part 1: Principles, Requirements and Tests*, International Electrotechnical Commission, 2020, with Amendment 1, 2025. [Online]. Available: https://webstore.iec.ch/en/publication/59671
[84] A. Hopkins, B. Hopfensperger, and P. Mellor, "Dc-link capacitor reduction in low voltage and high power integrated modular motor drives," in *2019 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2019, pp. 3208–3214.
[85] S. Wang, M. Rivera, D. Golovanov, J. Blissett, J. Moran, P. Davison, G. Ramtharan, C. Gerada, and P. Wheeler, "A hybrid pwm technique employing variable switching sequences for current ripple suppression in dual three-phase pmsms," *IEEE Trans. on Ind. Electron.*, vol. 72, no. 5, pp. 4577–4587, 2025.
[86] S. Wang, M. Rivera, D. Golovanov, J. Blissett, J. Moran, P. Davison, R. G., C. Gerada, and P. Wheeler, "Investigation of pwm techniques for high-frequency harmonics loss performance in dual-three phase asymmetrical machines," *IEEE Trans. on Ind. Electron.*, vol. 72, no. 8, pp. 7826–7838, 2025.
[87] J. Yang, G. C. Lim, and J.-I. Ha, "Hybrid modulation for zero-sequence current suppression in common-source fed open-end winding motor drive system," *IEEE Trans. on Power Electron.*, vol. 41, no. 9, pp. 16 385–16 399, 2026.
[88] G. Lee, M. Kim, H. Kim, M. Kim, G. Choi, T. Chung, H. Park, S. Han, M. Kim, and K. Lee, "Alternating-sequence discontinuous pwm for open-end winding pmsm fed by dual inverter with a common dc bus," *IEEE Trans. on Ind. Electron.*, pp. 1–12, 2026.
[89] Y. Sun, Z. Wang, Y. Shen, Z. Xu, H. Chang, H. Zhu, and H. Xiong, "Mitigation of common-mode voltage in 3l-vsi-fed dual three-phase pmsm drives using model predictive control," *IEEE Trans. on Transp. Electrification*, vol. 11, no. 4, pp. 9439–9450, 2025.
[90] J. Zheng, S. Yang, Z. Xie, X. Zhang, and L. Chang, "Multiobjective pwm strategy with independent zsc suppression, cmv variation elimination, and npv balance for oewim fed by dual three-level vsi," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 13, no. 6, pp. 8017–8028, 2025.
[91] F. Wu, A. M. El-Refaie, and T. M. Jahns, "Active voltage balancing of integrated modular drive with series dc-link capacitors," in *2020 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2020, pp. 1469–1476.
[92] J. Wang, "Design of multilevel integrated modular motor drive with gallium nitride power devices," Ph.D. dissertation, University of Wisconsin-Madison, 2015.
[93] A. Demarcos, E. Robles, U. Ugalde, I. Elosegui, and J. Andreu, "Output current ripple in electric vehicles with asymmetrical dual three-phase arrangement applying interleaved carrier-based pwm techniques," *IEEE Access*, vol. 13, pp. 41 743–41 767, 2025.
[94] A. H. Mohamed, H. Vansompel, and P. Sergeant, "A generic dc link capacitor sizing methodology for multi-phase wide bandgap based integrated modular motor drives," in *2020 IEEE International Conference on Environment and Electrical Engineering and 2020 IEEE Industrial and Commercial Power Systems Europe (EEEIC / I&CPS Europe)*, 2020.
[95] C. Alosa, F. Immovilli, and E. Lorenzani, "Modular multi-three-phase electric drives for enhanced reliability and current ripple minimization," in *IECON 2019 - 45th Annual Conference of the IEEE Industrial Electronics Society*, vol. 1, 2019, pp. 7108–7114.
[96] T. Chen, C. Caicedo-Narvaez, H. Wang, M. Moallem, B. Fahimi, and M. Kiani, "Electromagnetic compatibility analysis of an induction motor drive with integrated power converter," *IEEE Trans. on Magn.*, vol. 56, no. 1, 2020.
[97] F. Giardine, S. Krishnan, M. Nerenberg, and R. C. N. Pilawa-Podgurski, "An analytical model for common-mode emi in flying capacitor multilevel converters," in *2025 IEEE 26th Workshop on Control and Modeling for Power Electronics (COMPEL)*, 2025, pp. 1–7.
[98] Z. Zhong, Z. Zhu, J. Sun, and H. Chen, "Modeling and suppression of common-mode current resonance in an all-in-one motor drive module of electric vehicle," *IEEE Trans. on Electromagn. Compat.*, vol. 67, no. 2, pp. 477–486, 2025.
[99] D. Han, W. Lee, S. Li, and B. Sarlioglu, "New method for common mode voltage cancellation in motor drives: Concept, realization, and asymmetry influence," *IEEE Trans. on Power Electron.*, vol. 33, no. 2, pp. 1188–1201, 2018.
[100] S. J. Lee, J. Choi, G. C. Lim, S. M. Kim, E.-K. Kim, and J.-I. Ha, "Analysis and mitigation of bearing current in inverter-fed ipmsms drive system," *IEEE Trans. on Ind. Electron.*, vol. 72, no. 4, pp. 3503–3513, 2025.
[101] Z. Li, R. Liu, Y. Hao, D. Zheng, and J. Gui, "A novel nonintrusive measurement method of bearing currents in inverter-fed motors," *IEEE Sensors J.*, vol. 25, no. 19, pp. 36 580–36 593, 2025.
[102] A. K. Khiz and S. H. Jayaram, "Effects of rise time and overshoot of wbg-based power converters on hairpin winding with corona-resistant and non-corona-resistant wires in evs," *IEEE Trans. on Dielectr. Electr. Insulation*, vol. 32, no. 4, pp. 2334–2343, 2025.
[103] M. Güleç, M. Azeem, K. B. Tawfiq, T. Helmholdt-Zhu, and P. Sergeant, "Toward accurate partial discharge measurements under impulsive pulse voltages for turn-to-turn insulation of inverter-fed motors: Time interval selection, frequency investigation, and pd pattern," *IEEE Access*, vol. 13, pp. 216 684–216 693, 2025.
[104] P. Seri, I. Iannarelli, F. Zecchin, A. Cavallini, and H. Ishikawa, "Influence of automatic transmission fluid on partial discharge inception voltage in hairpin stators for electric vehicles," in *2025 IEEE Conference on Electrical Insulation and Dielectric Phenomena (CEIDP)*, 2025, pp. 756–759.
[105] J. Kim, S. Shin, B. Battulga, M. F. Shaikh, Y. Park, C. Lim, and S. B. Lee, "Frequency response analysis for quality assurance testing of hairpin stator winding insulation for electric vehicle applications," *IEEE Trans. on Energy Convers.*, vol. 40, no. 1, pp. 690–693, 2025.
[106] I. Martinez, I. Iannarelli, P. Seri, and A. Cavallini, "Insulation technologies for state-of-the-art electrical machines used in transport electrification," in *2025 IEEE Workshop on Electrical Machines Design, Control and Diagnosis (WEMDCD)*, 2025, pp. 1–4.
[107] A. Ihsan, M. Z. Youssef, and S. S. Williamson, "Active emi filter in gan-based inverter application," in *2025 IEEE Energy Conversion Conference Congress and Exposition (ECCE)*, 2025, pp. 1–6.
[108] L. Xie, Y. Wu, Y. Li, W. Zhou, X. Yuan, and X. Ruan, "A cancellation-based circuit for reducing both the shaft voltage and cm noise in motor drive systems," *IEEE Trans. on Ind. Electron.*, vol. 73, no. 4, pp. 5223–5233, 2026.
[109] S. Ni, Z. Zheng, K. Liu, and L. Xu, "Stability analysis and voltage controller design for novel series-connected dc-link topology for mvdc multiphase drives," in *2025 IEEE 3rd International Power Electronics and Application Symposium (PEAS)*, 2025, pp. 2056–2060.
[110] A. Galassini, A. Costabeber, C. Gerada, G. Buticchi, and D. Barater, "A modular speed-drooped system for high reliability integrated modular motor drives," *IEEE Trans. on Ind. Appl.*, 2016.
[111] L. Zhu, Y. Yan, and H. Bai, "Design and transient analysis of a 650 v/150 a gan power modules with integrated bias power and gate-drive circuit," *IEEE Trans. on Components, Packag. Manuf. Technol.*, vol. 14, no. 3, pp. 417–427, 2024.
[112] M. D. Rahman, T. Akter, A. S. Md Khalid Hasan, Hamood-Ur-Rehman, Y. Chen, H. A. Mantooth, and X. Song, "Design and evaluation of a 650 v, 300 a gan-based power module with integrated drivers and ultra-low inductance layout," in *2025 IEEE 12th Workshop on Wide Bandgap Power Devices and Applications (WiPDA)*, 2025, pp. 1–5.
[113] J. Ajayan, A. Kumar Panigrahy, S. Sen, M. Kumar, and S. Tayal, "Reliability issues and degradation mechanisms of p-gan gated e-mode algan/gan power hemts: A critical review," *IEEE Access*, vol. 13, pp. 84 541–84 554, 2025.

[114] J. Ajayan, V. B. Sreenivasulu, N. A. Kumari, S. Sreejith, and S. Das, "Challenges and advances in materials and fabrication technologies for the development of p-gan gated e-mode algan/gan power hemts: A critical review," *IEEE Access*, vol. 13, pp. 62 045–62 059, 2025.

[115] M. Nikouie, O. Wallmark, L. Harnefors, and H.-P. Nee, "Operation under fault conditions of the stacked polyphase bridges converter," in *IECON 2016 - 42nd Annual Conference of the IEEE Industrial Electronics Society*, 2016, pp. 2207–2211.

[116] H. Zhang, O. Wallmark, and M. Leksell, "On fault tolerance for IPM-FSCW machines adopting a modular converter," in *2014 17th International Conference on Electrical Machines and Systems (ICEMS)*, 2014, pp. 1633–1638.

[117] S. S. Gjerde, P. K. Olsen, K. Ljokelsoy, and T. M. Undeland, "Control and fault handling in a modular series-connected converter for a transformerless 100 kv low-weight offshore wind turbine," *IEEE Trans. on Ind. Appl.*, vol. 50, no. 2, pp. 1094–1105, 2014.

[118] X. Zhang, P. Tian, K. Chen, A. T. Parra, T. M. Jahns, and B. Sarlioglu, "Development and testing of a fault-tolerant modular motor drive using current-source inverters," in *2025 IEEE Energy Conversion Conference Congress and Exposition (ECCE)*, 2025, pp. 1–7.

[119] Y. Du, Y. Li, K. Wang, Z. Zheng, and C. Gerada, "Dc injection based fault-tolerant control of modular pmsm drives under single/hybrid open-switch faults," *IEEE Trans. on Ind. Electron.*, vol. 73, no. 5, pp. 6646–6656, 2026.

[120] A. Ebrahimian, S. I. H. Sabzevari, and N. Weise, "Control & communication architecture for a fault-tolerant integrated modular motor drive," in *2025 IEEE Energy Conversion Conference Congress and Exposition (ECCE)*, 2025, pp. 1–7.

[121] J. A. Swanke and T. M. Jahns, "Reliability analysis of a fault-tolerant integrated modular motor drive for an urban air mobility aircraft using markov chains," *IEEE Trans. on Transp. Electrification*, vol. 8, no. 4, pp. 4523–4533, 2022.

[122] Efficient Power Conversion, *EPC2361 Enhancement Mode Power Transistor*, Efficient Power Conversion, 2026, datasheet, rev. July 20, 2026.